

Zhihan Yin

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RESEARCH INTERESTS

Multimodal agents; autonomous cross-modal search; vision-language model reasoning; fine-grained visual perception and hallucination; alignment and RL post-training.

EDUCATION

Peking University | Aug 2023 - Present

B.S. Candidate, Intelligence Science and Technology | Cumulative GPA: 3.69 / 4.00

SELECTED PUBLICATIONS AND MANUSCRIPTS

FREAK: A Fine-grained Hallucination Evaluation Benchmark for Advanced MLLMs

Zhihan Yin, Jianxin Liang, Yueqian Wang, Yifeng Yao, Huishuai Zhang, Dongyan Zhao

ICLR 2026

A benchmark built from photorealistic images with fine-grained counter-commonsense edits, probing whether advanced multimodal models perceive visual details or fall back on familiar priors.

Veto: A Training-free Framework for Fine-grained Visual Perception via Iterative Negative Filtering

Ruyi Li, **Zhihan Yin**, Tan Yue, Jianxin Liang, Xian-Feng Han, Huishuai Zhang, Dongyan Zhao

ACM Multimedia 2026

A training-free perception framework that progressively removes distracting image regions, helping vision-language models locate and verify fine-grained visual evidence.

MIRA: Reinforcing Multimodal Reasoning via Deceptive Contextual Augmentation

Zhihan Yin, Jianxin Liang, Yifeng Yao, Nonghai Zhang, Bingyue Peng, Chunguang Qie, Huishuai Zhang, Dongyan Zhao

NeurIPS 2026 submission

Reinforcement learning with deceptive visual contexts and group-wise reflection triggers, encouraging multimodal models to revisit perceptual mistakes and recover during reasoning.

RESEARCH AND TEACHING EXPERIENCE

Research Intern, Monetization GenAI, ByteDance

Apr 2026 - Present

Research on multimodal agents.

Research Intern, Wangxuan Institute of Computer Technology, Peking University

Mar 2025 - Present

Research on multimodal large language models, visual perception, hallucination evaluation, and reasoning.

Teaching Assistant, Introduction to Artificial Intelligence, Peking University

Mar 2026 - Present

Course instruction support, assignment design and grading, and student mentoring.

HONORS AND AWARDS

Merit Student (Top 10% at Peking University) | 2025, 2026

Huirong Li Scholarship (Top 15% at Peking University) | 2025

National Encouragement Scholarship (Top 30% at Peking University) | 2024, 2025

SKILLS

Programming: Python, C++

Deep learning: PyTorch, vLLM, veRL

Research tools: LaTeX, Adobe Illustrator

English: CET-4: 580; CET-6: 570